

KNOWLEDGE TRANSFER QUESTIONNAIRE

Purpose: When teammates depart from State of Nebraska, they often carry institutional knowledge that helped them be successful. This tool helps the State of Nebraska retain this knowledge for business continuity.

How-to: Help us capture information about your job that might not be written down elsewhere. Please follow the prompts below. Attach additional pages or materials if needed.

TEAMMATE NAME: Christopher Austin	AGENCY & DIVISION: DHHS Public Health - Epidemiology
JOB TITLE: Occupational Health Epidemiologist	SUPERVISOR: Derry Stover
1: MANY TIMES, WE DEVELOP "TRICKS OF THE TRADE" TO MAKE US MORE ACCURATE AND EFFICIENT. WHAT ARE THE TOP 5 (OR MORE) LESSONS OR "TRICKS OF THE TRADE" THAT YOU'VE LEARNED IN YOUR PRESENT POSITION?	
1) Automating Data Retrieval for Efficiency: one 'trick' I've learned is automating data retrieval using APIs. For example, I developed to pull work fatality data from the Bureau of Labor Statistics (BLS) API saves considerable time by batching requests, handling large data sets, and efficiently querying fatality information. By structuring the code to handle multiple queries and split datasets automatically, I reduced the manual effort involved in retrieving data, ensuring the process is both scalable and efficient. This approach can be applied to any API-based data retrieval. 2) Using Web-Based Tools for I/O Standardization: Using web-based tools like the NIOCCS auto-coder for translating free-text industry and occupation data into standardized codes (NAICS, SOC, Census) is an essential trick for ensuring consistency. Automating this process through API calls not only eliminates manual coding errors but also allows large datasets to be processed quickly. This technique is particularly useful when working with occupational health surveillance data, ensuring that data analysis is based on consistent classifications. 3) Applying Natural Language Processing (NLP) for Text Data: Processing and analyzing unstructured text data using NLP techniques has been crucial in improving the accuracy of work-related injury classifications. Techniques like tokenization, stop word removal, and lemmatization help clean and standardize the text data. Additionally, feature extraction using methods like TF-IDF or Count Vectorization feeds the processed text into machine learning models, ensuring that even free-text fields are handled in a structured, reproducible manner. This makes analyzing large volumes of text data more efficient.	
2: WHICH PROCEDURES (EXAMPLES: NAMES OF MANUALS, REFERENCES, ETC.) DO YOU TURN TO THE MOST TO DO YOUR JOB?	
Most procedures I turned to are one I created or developed for my work (see document for these details). However, things I have turned to that helped me are R package cheat sheets (see Q3), occupational health for syndromic surveillance white paper, NAICS information (https://www.census.gov/naics/), NIOCCS material (https://www.cdc.gov/niosh/topics/coding/nioccs/support.html), and NIOCCS autocoder (https://csams.cdc.gov/nioccs/About.aspx). I also refer to OSHA codes and relations to help understand employer expectations to provide a safe working environment, https://www.osha.gov/laws-regs, and how injuries might be recorded per OSHA regulations.	

3: IF YOU HAD JUST ONE DAY TO TRAIN OR BRIEF YOUR REPLACEMENT, WHAT 5 (OR MORE) THINGS WOULD BE MOST IMPORTANT TO SHARE?

- 1) NIOCCS Auto-Coding: Another critical tool is the NIOCCS Auto-coder, which I developed to streamline the process of converting free-text industry and occupation data into standardized codes using the NIOSH Industry and Occupation Computerized Coding System (NIOCCS). This tool automates the web API calls to NIOCCS, processes large datasets efficiently, and returns standardized NAICS and SOC codes. It's vital to know how to properly structure the API queries and handle the responses to ensure accurate coding of occupational data, which is essential for occupational health studies and surveillance.
- 2) Workforce Estimates Calculations: One of the key tools I developed is the Workforce Estimates Calculation Tool, which automates the process of extracting and calculating employment estimates from ACS PUMS data. This tool also adjusts for Full-Time Equivalency (FTE) by considering the hours worked per week, providing a more accurate representation of workforce sizes in various industries. It's essential to understand how to extract the relevant variables, process the data using svyr, and calculate employment estimates while handling survey weights. I would walk through how to ensure precision in the results by calculating margins of error (MOE), relative MOE, and coefficients of variation (CV), which are critical for ensuring data quality and reliability.
- 3) Data Processing Workflow: A step-by-step guide on how to preprocess and analyze syndromic surveillance data, particularly for occupational injuries. This includes running scripts in R for data cleaning and quality checks, as well as ensuring accurate categorization of work-related incidents.
- 4) Key Machine Learning Models: How to use and interpret the machine learning models I've developed, particularly the Logistic Regression model, which has been the most effective for classifying work-related injuries. I would provide documentation on model parameters, data preprocessing, and troubleshooting tips.
- 5) Ongoing Projects and Priorities: Details on any ongoing occupational health projects (e.g., syndromic surveillance occupation health query and occupational health tribal guidance document), including deadlines, key goals, and next steps. This ensures that they can hit the ground running on important work.
- 6) Documentation and Resources: Where to find key documentation, and cheat sheets. This would include access to shared folders with important reference materials and guides.

4: WHERE DO YOU KEEP OR STORE THE FOLLOWING ITEMS?

Electronic files, records, and/or data for team use:

K:\Occupational Health Grant\Chris Austin

Electronic files, records, and/or data for your use:

K:\Occupational Health Grant\Chris Austin

Hardcopies, paper files, and other physical records:

In file cabinet in cubicle 125

Anything else?

5: DESCRIBE ANY TOOLS THAT YOU CREATED TO HELP YOU PERFORM YOUR JOB MORE EFFICIENTLY. EXAMPLES COULD INCLUDE HOW-TO'S, "CHEAT SHEETS," JOB AIDS, WORKFLOWS, CHECKLISTS, CALENDARS, AND THE LIKE.

NIOCCS Auto-Coder: I developed an automated tool in R to convert free-text industry and occupation (I/O) data into standardized codes using the NIOSH Industry and Occupation Computerized Coding System (NIOCCS). This tool leverages the NIOCCS Web API to automatically code raw text into standardized NAICS, SOC, and Census codes, eliminating the need for manual coding. This is especially useful for large datasets, such as COVID-19 surveillance data that includes industry and occupation information. The tool not only speeds up the process but also ensures coding accuracy and consistency across datasets. It integrates seamlessly into data processing pipelines and has proven to be essential in occupational health surveillance.

Workforce Estimates (FTE) Calculation Tool: I developed an R-based tool to calculate workforce estimates from the American Community Survey (ACS) Public Use Microdata Sample (PUMS) data and convert these estimates to Full-Time Equivalent (FTE) rates. This tool automates the extraction of relevant variables from the ACS PUMS data (e.g., employment status, industry codes, hours worked per week) and applies statistical methods to calculate employment estimates by industry.

6: LIST ANY JOB FUNCTION(S) WHERE YOU ARE THE ONLY PERSON WHO KNOWS HOW TO GET IT DONE.

If someone is proficient in R they should be able to understand and run the scripts I created during my job as Occ Health Epi.

Performing NLP and machine learning to identify work-related injuries in syndromic data might be an area that not many people are familiar with. But notes and documentation is provided to help guide anyone interesting in pursuing this further.

7: WHO ARE THE PEOPLE YOU INTERACT WITH THE MOST WITHIN YOUR ORGANIZATION IN RELATION TO YOUR JOB DUTIES?

Name	Position	Contact Info	Reason	Frequency
Derry Stover	Epidemiology Surveillan	derry.stover@nebraska	Supervisor	Weekly

8: WHO ARE THE PEOPLE YOU INTERACT WITH THE MOST OUTSIDE YOUR ORGANIZATION? THESE COULD INCLUDE CONTACTS AT LOCAL, STATE, OR FEDERAL LEVELS.

[illegible]

9: DO YOU HAVE ANY ONGOING PROJECTS THAT SOMEONE ELSE SHOULD TAKE ON?

Project/Activity	Project Teammates	Is Anything Due Soon?	Project Due Date	Who else could take this on?

10: PROVIDE THE FOLLOWING ACCESS INFORMATION. WHERE NECESSARY, USE DISCRETION AND PROVIDE TO AUTHORIZED PERSONNEL ONLY.

Desk phone PIN for voicemail:

State cellphone PIN(s) and passwords for voicemail and/or to unlock phone:

Combination to safe, locker, etc.:

Access information for other devices (list device with access information):

11: WHERE PERMISSIBLE, PROVIDE LOGIN INFORMATION FOR ONLINE ACCOUNTS FOR OUTSIDE VENDORS AND SYSTEMS NOT MANAGED BY OCIO (EXAMPLES: ENVIRONMENTAL MONITORING SYSTEM; FEDERAL REPORTING/TRACKING SYSTEM; ONLINE COMMUNITY FOR COLLABORATION).

Vendor/Website/System	Username	Password	Who knows how to use it?

12: IF YOU ARE RESPONSIBLE FOR CONFIDENTIAL MATERIALS OR INFORMATION, ARRANGE FOR THE PROPER TRANSFER OR DISPOSITION OF THESE MATERIALS BEFORE YOUR DEPARTURE. WHERE ALLOWABLE, PROVIDE INSTRUCTIONS, LOCATIONS, OR RELATED INFORMATION BELOW.

Any confidential material will stay securely on internal drives and folders controlled by NDHHS. Relevant folder locations and files are provided above.

13: IF YOU HAVE AN ONLINE VENDOR ACCOUNT WITH OFFICE DEPOT, FISHER SCIENTIFIC, GRAINGER, MSC, FASTENAL, CARDINAL PHARMACEUTICAL, VWR, ETC., WHO WILL CONTINUE PLACING AND/OR APPROVING ORDERS AFTER YOU LEAVE.

Not Applicable

14. ARE THERE ANY FUNCTIONS, WITH UPCOMING OR RECURRING DEADLINES, THAT WE NEED TO BE AWARE OF? (EXAMPLES: GRANT RENEWALS, PROGRESS REPORTS, SERVICE CONTRACT RENEWALS, PROFESSIONAL MEMBERSHIPS, ANNUAL REGISTRATION RENEWALS, ETC.)

What will become due?	Due date

15: IS THERE OTHER INFORMATION ABOUT YOUR WORK/POSITION THAT IS IMPORTANT TO CAPTURE FOR THE PURPOSE OF ONBOARDING A SUCCESSOR OR MAINTAINING ORGANIZATIONAL KNOWLEDGE?

Data Management and Access: A significant part of this role involves managing multiple data sources, including syndromic surveillance data, ACS PUMS data, and occupation/industry coding techniques, BLS data, Workers Compensation. It's important to ensure continued access to these datasets by maintaining any necessary credentials for tools for API access, tidycensus for PUMS data, and any secured databases within the state health department. Documentation on how to access these resources and the protocols for securely handling sensitive data should be well-maintained and shared with the successor.

Additional Thoughts on Workforce Estimates and FTE Calculations: The workforce estimates and FTE adjustment protocols I developed is a core resource for occupational health surveillance. This method requires an understanding of how to use survey weights from ACS PUMS data and how to interpret employment estimates both in their raw form and when adjusted for FTE for accurate and consistent workforce analysis.

Thank you for sharing your knowledge and expertise! Your contributions here will help us continue to Grow Nebraska!